Exercise 5: Multi-Criteria Evaluation 2

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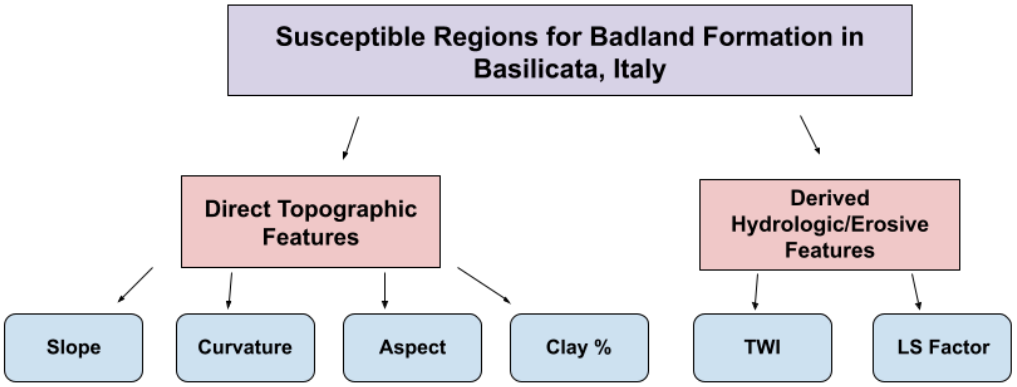
AHP DELIVERABLES:

1.



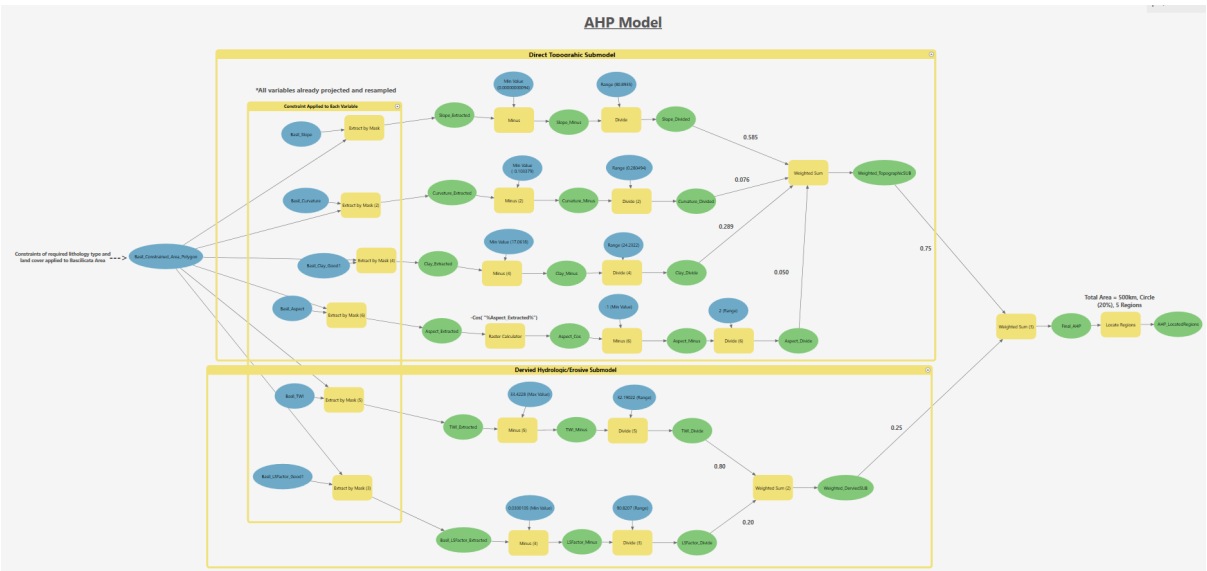
I chose to structure the decision problem in this way for a few reasons. To begin with, I chose two main groupings named "direct topographic features" and "derived hydrologic/erosive features" respectively. Direct topographic features refers to factors obtained from only one tool or calculation with no further processing required; slope, curvature, aspect, and clay % fit into that category. Derived hydrologic/erosive features refers to factors obtained using more complex calculations and multiple tools; TWI as well as LS factor fit into that category. LS factor incorporates slope steepness as well as slope length, while TWI incorporates even more factors such as flow accumulation and the slope surface parameter amongst a few others. I think it makes sense to separate the direct and derived features since different levels of processing are required for both and they serve different levels of importance. For example, since TWI and LS factor incorporate slope into their formulas I do not think it makes the most sense to compare them with the slope variable (in pairwise comparisons).

2. In order to standardize the layers I used a score range transformation on each variable. The reason I used the score range procedure instead of using the maximum range procedure was due to the fact that some of my input variables consisted of negative values. For example, the curvature data had negative values, which the maximum score procedure is not fit for. Also, by using the score range procedure you can easily use the cost and benefit version in the same model, but with the maximum score procedure you have to inverse either the cost or benefit equation so that the denominators are the same and then you can use it. For slope, curvature, clay percent, and LS factor I used the benefit criteria version of the score range procedure since those are the variables that we want a high score for. In order to apply the benefit

version of the score range procedure I first subtracted each layer from the minimum value found in each respective layer, then divided it by the range of values (max minus min) found in the original layer. For TWI I used the cost criteria version of the score range procedure since we want a low value of TWI. In order to apply the cost version of the score range procedure I first subtracted the maximum value found in each layer to each non-transformed layer, then divided it by the range of values found in the original layers. For aspect, since a value of 180 degrees is desired, but the range of values is from -1 to 360, neither benefit or cost range procedures would work. However, by applying a negative cosine function, 180 degrees would now give a peak value of “1”, while values further away from 180 degrees such as 360 degrees would give a value of “-1”. After applying this transformation the updated aspect layer can now be further transformed using the benefit version of the score range procedure. Finally, all layers now have data values ranging from 1 through 0 (based on importance) and they can now be combined using the weighted sum tool.

3. See attached spreadsheet.

4.



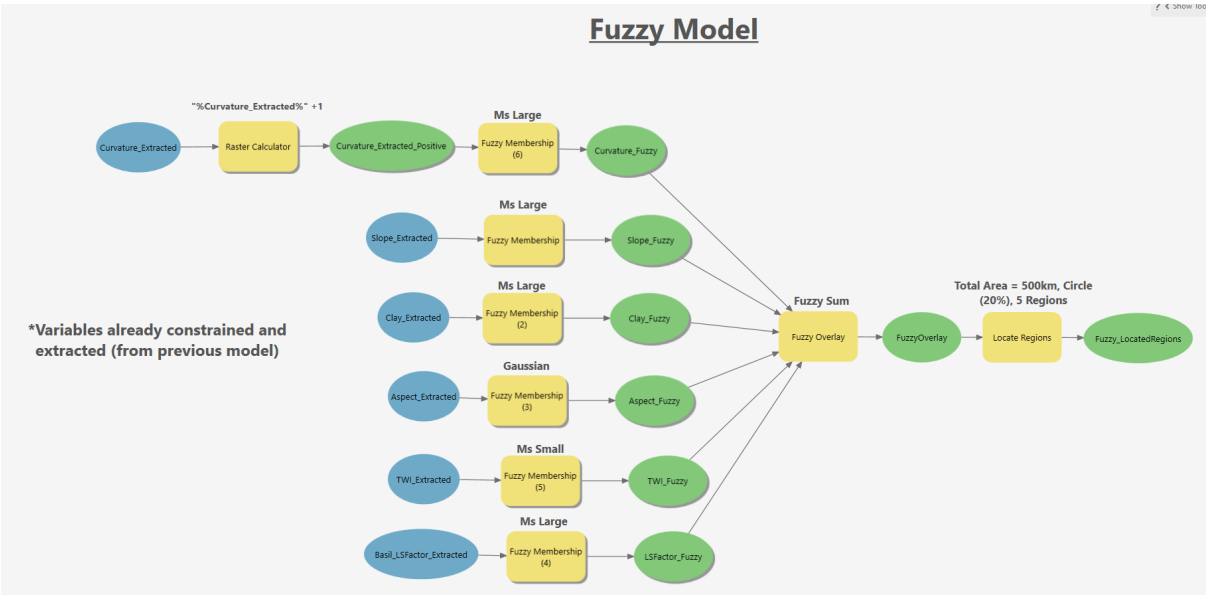
5. See AHP map.

6. In order to fuzzify the layers various fuzzy memberships were applied. Before applying the fuzzy memberships each layer was resampled, projected, and constraints were applied extracting only the desirable areas. For slope, clay, LS factor, and curvature the fuzzy

membership used was the fuzzy MS large. The MS stands for mean and standard deviation, which is what that membership type is based on. The reason why I used this type of membership is due to the fact that for those factors a larger value is favoured. However, the fuzzy MS large function does not work for negative values, so for the curvature variable which initially had a range of values into the negatives a transformation had to be applied. This transformation included adding a value of "1" to all of the original curvature values so that all the values are now positive and the MS large function can now be used. For TWI, the fuzzy MS small function was used, since we want smaller values to be more likely to be a part of the set. Finally, for aspect the fuzzy Gaussian function was used. Unlike the AHP model, the aspect variable did not have to be transformed before further processing was done. As such, the original range of values of -1 through 360 can be used. A midpoint value of 180 was set, since a southerly aspect direction is preferred. The fuzzy Gaussian function was used instead of the fuzzy near function because the Gaussian function allows for a larger spread of values while the near function has a thinner spread, and since a specific value of 180 is not 100% required, only a southerly direction is preferred, that means a more gradual approach can be used. Once all of the layers were properly "fuzzified", each layer now had values ranging from 0 through 1 and a fuzzy overlay can be applied. The type of fuzzy overlay I used was the fuzzy sum. I used this type because by applying it to the already fuzzified layers it allows us to find all *possible suitable* locations for badland formation. I could have also used the fuzzy product overlay since this function is best used for identifying the *best* locations. However, I used the locate regions tool after the fuzzy sum overlay to give us the five best regions for badland formation.

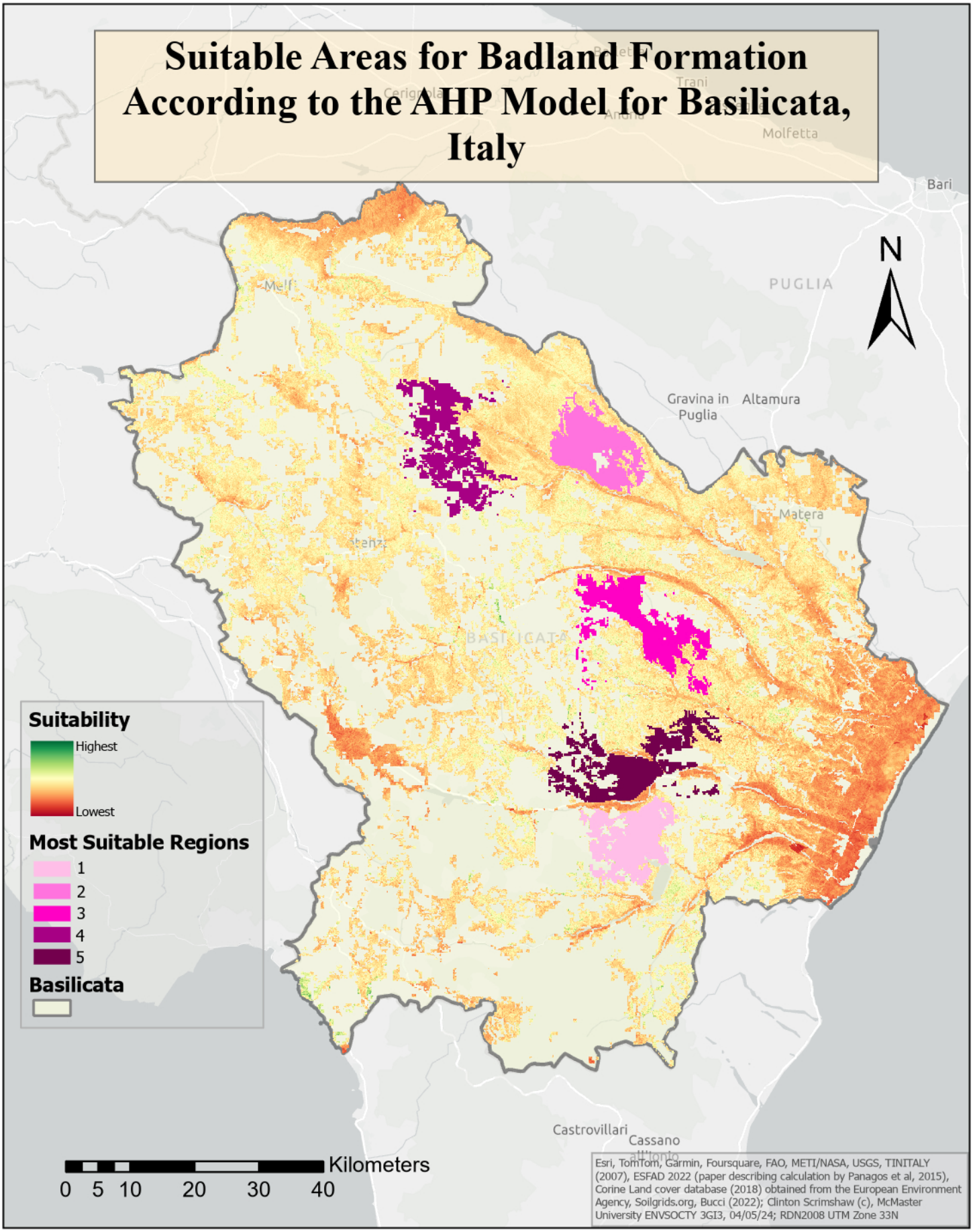
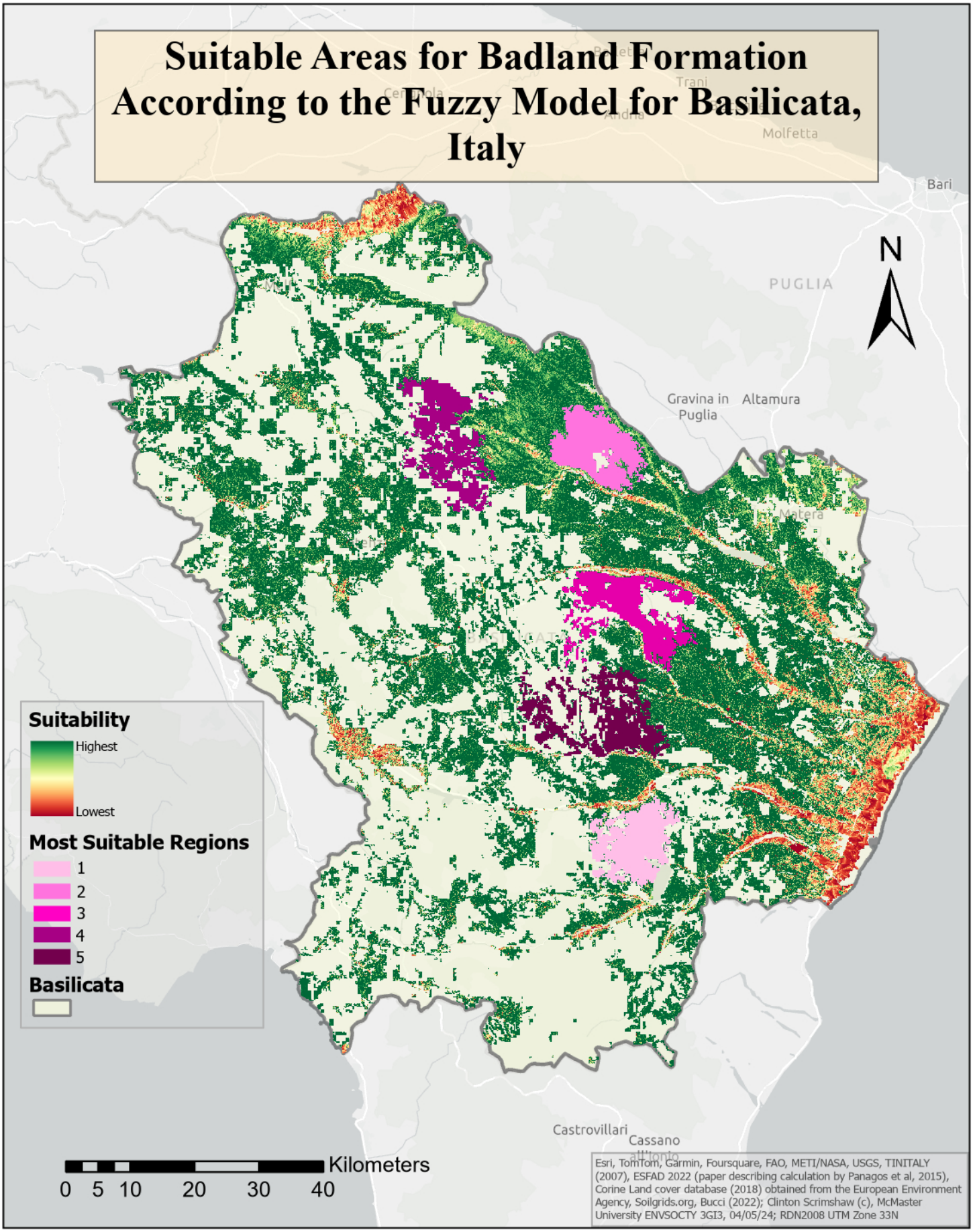
In conclusion, by applying the fuzzy memberships and the fuzzy overlay it allows for uncertainty and ambiguity to be better represented in the output layer. Instead of using crisp cut-off values, fuzzifying the data allows for errors to be taken into account. This is beneficial for the cases where measurement or calculation errors cause data points to be excluded when they may actually be of importance.

7.



8. See fuzzy map.

Suitable Areas for Badland Formation According to the AHP Model for Basilicata, Italy



Suitable Areas for Badland Formation According to the Fuzzy Model for Basilicata, Italy